

## Calculus 1 sec 2: Assignment 2, due Tuesday Feb 17, 2026, 11:59pm

Submit the completed assignment to <https://www.gradescope.com>.

Only hand-written solutions are accepted.

- This assignment is worth a total of 15 points, and accounts for about 3% of your final grade. The late submission deadline (no penalty) is Thursday, Feb 19, 11:59 pm. Submissions after the late due date will be penalized and NOT be accepted unless you have a good excuse.
- Collaboration is allowed, but you must write up your solutions independently and in your own words. Copying from others, publications, or websites (even with minor changes) will be treated as plagiarism.
- Neatness, clarity, and correct notation matter! Show all steps, use clear explanations, and write in complete sentences where needed. Graders will only evaluate what you write, so unclear or incomplete work will affect your grade, even with the correct answer.
- Attempt all questions (and parts), and clearly indicate which question you are answering. Answer each question on a separate sheet, and please **match your answers with the correct questions on Gradescope**.

There are 5 questions plus a bonus.

- 1 (i) [1.5 pt] Find the values of  $a$  such that the limit  $\lim_{x \rightarrow a} \frac{|x|-a}{x-a}$  exists.  
*Hint:* try this for different values of  $a$  and see what you find.

- (ii) [2 pt] Find the number  $c$  such that the limit

$$\lim_{x \rightarrow -1} \frac{3x^3 - 3x^2 + 2c - 2}{x^3 + 1}$$

exists. What is the value of the limit?

*Show all the steps*

2. [2 pts] Find the values  $a$  and  $b$  so that the function

$$f = \begin{cases} a(2-x)^2 + b - 1 & \text{if } x \leq 0 \\ 3ax \sin\left(\frac{\pi}{x}\right) & \text{if } 0 < x \leq 4 \\ a - 1 + \frac{b(x-4)}{2x^2 - x - 28} & \text{if } 4 < x \end{cases}$$

is continuous.

*Show all the steps*

3. Find the limit, if it exists. If the limit does not exist, explain why.

(a) [1.5 p]  $\lim_{x \rightarrow 1} \cos\left(\frac{1}{x-1}\right)$

(b) [1.5 p]  $\lim_{x \rightarrow 1} \frac{x-1}{\cos(x-1)}$

*Show all the steps.*

4. (a) [1p]

(i) State the Intermediate Value Theorem (IVT).

(ii) Show by means of an example that the conclusion of IVT may hold for a function that is continuous on an interval  $[a, b]$  except for at one point  $c \in (a, b)$ . Include the graph of such function with your solution.

(b) [2p] Use IVT to find all values of  $c$  such that the equation

$$\arcsin(x) = 2x - c$$

has a solution in the interval  $[0, 1]$ . Show all the steps.

5. Let  $f(x) = 3x - 1$  and  $h(x) = \frac{x+2}{x^2-x-2}$ .

(a) [1.5 pt] What are the domains of  $h$  and  $h \circ f$ ?

(b) [2 p] Find a formula for the inverse  $f^{-1}$  of  $f$  and use it to find a function  $g$  such that  $f \circ g = h$ .

Hint: remember that the functions  $f^{-1} \circ f$  and  $f \circ f^{-1}$  both equal the identity whenever they are defined.

*Show all your steps*

**Optional** (*extra credit, complete solutions only – no partial credit*)

**Bonus** [2pt] Consider the function  $f(x) = \ln(x + \sqrt{x^2 + 1})$ .

What is its domain?, its range? is it continuous? does it have any vertical asymptotes?

You must explain your answers using properties of the formula of the function.